Energy 258 (2022) 124741

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Future global electricity demand load curves

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ARTICLE INFO

Keywords:
Integrated model
Electricity demand
IAM
VRE
SSP
Scenarios

ABSTRACT

The rapidly increasing electricity demand and the expected increase in the contribution of variable renewable energy sources raise the need for looking at the characteristics of long-term demand variations. Furthermore, demand changes (e.g., the increasing penetration of electric vehicles) could affect the shape of future load curves. However, integrated assessment models often assume a constant load shape. In this research, the shape of future electricity demand load curves is determined with a global scope for long-term exploratory scenarios analysis within integrated assessment models. This was done by using empirical data on daily demand patterns of different end-uses and aggregating them with end-use annual electricity demand data from the IMAGE model. The regional hourly aggregated patterns modelled vary over the years by projected variation of temperature and contribution variations of the different sectors to total electricity demand. Results under the shared socioeconomic pathway two show that future load curves depict low changes over time and a large sensitivity to load variations from electric vehicle daily charging patterns.

1. Introduction

Global electricity demand is expected to grow rapidly in the next decades, driven by economic growth, structural changes, and further electrification [1]. As the growth is unevenly distributed between the different end-uses of electricity, not only is the total volume subject to change but also the demand variation over a day. This could have clear consequences for the design of future energy systems [2]. In particular, the variation of electricity demand over a day, month, and year can play a crucial role in enabling high shares of variable renewable energies (VRE, such as wind and solar power) into the grid and estimating the associated costs. Furthermore, it has been demonstrated that planning electricity systems with high penetration of VRE requires high time resolution models [3].

In recent years, dedicated energy models have improved the representation of short-term dynamics [4]. Chan et al. [4] surveyed fifty-four energy models, with most of them having an hourly resolution. Toktarova et al. [5] developed long-term load projections for all countries implementing multiple linear regression. Their projections depend on socioeconomic variables and consider the effect of some end-uses but exclude the expected influence of electric vehicles. The TIMES [6] and GENeSYS-MOD [7] models account for intra-annual variation with time-slices that differentiate day and night, weekdays and weekends, and seasons. Furthermore, they account for demand-side flexibility for heat and electricity. Another dedicated global model is Balmorel, which has an hourly resolution [8], and for Europe, a module was developed for analyzing end-use demand response to electricity prices [9]. Additionally, there are dedicated tools for analyzing demand load changes over time, such as Eload and Destinee [10], but they depend on other model data for a complete analysis of the energy system [11].

Future electricity systems can also be studied with integrated assessment models (IAMs). The latter are tools for understanding long term development pathways of global energy demand and supply under varied climate change mitigation strategies with the interaction of socioeconomic, biodiversity and land-use scenarios. However, due to their global, wide system scope and long-term time horizon, IAMs face important challenges when modelling short-term dynamics of the power sector [12]; still, these dynamics are important for future projections [13] under climate change mitigation scenarios. IAMs mostly implement load duration curves (LDCs) to capture short-term dynamics [14]. LDCs represent load demand sorted from the highest load to the lowest load. It enables analyzing the dispatch order of energy sources, and they allow,

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https://doi.org/10.1016/j.energy.2022.124741
Received 20 September 2021; Received in revised form 20 June 2022; Accepted 2 July 2022
Available online 10 July 2022

for instance, identifying the fraction of time with a high or low electricity demand. A steep LDC function will make it more difficult for renewable electricity to achieve high penetration levels in the energy system if the supply of electricity and peak demand are out-of-phase. However, currently, the shape of the LDCs is often assumed to remain constant and based on historical demand data [15].

The IAMs GCAM and POLES-JRC analyze some intra-annual demand variation. The first one captures load demand variations over time depending on temperature changes and considering day/night variations [16]. However, the results are given only for USA, and there is no illustration of the changes in the shape of load curves over time. POLES-JRC implements a representative summer and winter load curve per sector with a global scope [17]. Furthermore, for the Europe region, POLES-JRC is coupled with a short-term dispatch model (EUCAD) for analyzing electricity storage and demand response options [18]. PRIMES is a regional IAM with a European scope that models intra-annual demand variability depending on individual load profiles of end-uses [19].

A recent improvement for global IAMs is described in the paper of M. Brinkerink et al. [20], in which scenario data obtained with the IAM MESSAGE is implemented in the high temporal resolution power system model PLEXOS such as its results can be used to update parameters in MESSAGE (e.g. capacity factors, curtailment values and generation cost). Although the proposed methodology in that paper can improve the accuracy of IAMs results, the high-resolution PLEXOS model only focuses on the power sector. Therefore, it cannot model demand for hydrogen from other sectors (such as transport) or changes in the shape of demand due to structural changes such as the increased use of electric vehicles, heat pumps or air cooling devices. Given this gap, this research aims to improve the representation of energy demand short-term dynamics within IAMs. To this end, time-dependent hourly demand patterns are developed for large global regions and associated LDCs based on projected structural changes in demand, empirically observed demand patterns per end-use, and stylized functions. This is done by using the projections of the IAM model IMAGE on electricity demand by end-use function (e.g. lighting or heating) [21]. These demand patterns that change over time can subsequently be used in IAMs.

This paper is organized as follows. Section 2 introduces the methodology for modelling hourly electricity demand patterns. In Section 3, the resulting regional demand patterns are compared with their analogue empirical data to assess the model performance; and the projected changes over time are described. Finally, Section 4 presents the discussion and conclusions.

2. Methodology

The method presented here looks at the typical demand pattern per end-use function (e.g. electricity use for lighting is larger during dark evening hours). The residential and service sectors are divided into six end-uses: lighting, cooling, heating, water heating, cooking, and appliances/equipment. Demand in the industry sector is broken down into four categories: food, paper, cement and steel. Furthermore, the transport sector is divided into electric vehicles (EV) and railways. Energy demand from other sectors (non-specified demand, agriculture, forestry and fisheries) and other industries are assumed to be one function. The overall structure of the methodology implemented is schematically shown in Fig. 1. Future demand on an hourly basis is calculated using annual projections (obtained from IMAGE model) and end-use hourly load demand patterns based on empirically-based data found in the literature. The hourly patterns are related to metrics that influence the daily and monthly shape of end-use demand curves in order to be able to extrapolate the scarce available data to all regions. The metrics are building occupancy profiles and biophysical parameters for those end-uses found to have climate dependence. The resulting hourly electricity demand curves can subsequently be validated against empirical regional electricity demand data on an hourly basis.

HED stands for hourly electricity demand for each end-use e, which varies over regions r, months m, hours h and year y. *HPatt* stands for hourly patterns, and *YED* stands for yearly electricity demand. This means that for each end-use or sector, information is needed on 1) the variation of demand per month over the year and 2) the variation of demand over a day. For the latter, two typical hourly day patterns were used, one for working days and another for weekend days, following the variations already identified in total electricity demand [22]. The methodology applied is in line with the engineering method [23]. For all end-uses/sectors, the hourly pattern (*HPatt*) was calculated by multiplying the daily variation (*DV*), which varies for weekdays or weekends, and monthly variation (*MV*), as shown in Equation (2). Hence 24 day-demand-patterns are obtained for each end-use.

$$HPatt_{r,e,m,h} = DV_{r,e,m,h} * \ MV_{r,e,m} \tag{2}$$

The daily variation (*DV*) is calculated as shown in Equation (3) for end-uses in the residential and service sectors identified as a relationship between climate conditions and electricity demand for the monthly and hourly aggregating level (listed in Table 1). For each of these end-uses, a specific stylized function (f_e) was formulated that depends on building hourly occupancy profiles (*Occ*) and on the hourly and monthly

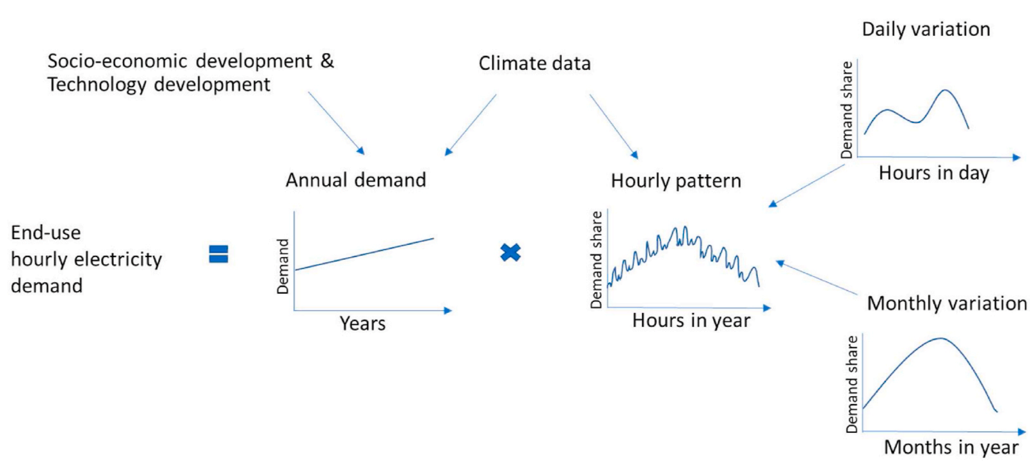

Fig. 1. Schema of the overall structure of the model.

$$HED_{r,e,y,m,h} = YED_{r,e,y} * \ HPatt_{r,e,m,h} \tag{1}$$

Table 1
Parameters that influence the modelled end-use patterns in the Residential and service sector (BP).

End-use	Biophysical parameter		Building occupancy profile dependence
	Monthly dependence	Hourly dependence	
Residential – Lighting	Irradiance	Irradiance	yes
Service – Lighting	Irradiance	–	–
Residential – Heating	Temperature	Temperature	yes
Service – Heating	Temperature	–	–
Residential & service- Cooling	Temperature	Temperature	yes
Residential & service – Hot Water	Temperature	–	–

variation of a biophysical parameter (*BP*) which are temperature or irradiance. Furthermore, some end-uses, such as water heating and lighting for the service sector, were identified to depend on climate conditions but only for monthly variation. In contrast, for their daily variation, an hourly pattern based on empirical data was implemented. In the end, the implementation of a climate parameter at the monthly and/or hourly aggregating level enables to have a residential and service sector pattern that varies across regions and over time.

$$DV_{r,e,m,h} = f_e(Occ_h, BP_{r,m,h}) \tag{3}$$

For end-uses without climate dependence in the residential and service sector, average weekend and weekday hourly patterns were determined from empirical data and literature research. An annual pattern describing monthly variations was applied to the end-use patterns that depicted a seasonal variation. A detailed methodology with the specific stylized functions for the end-uses with dependence on the biophysical parameters and the patterns implemented for end-uses without these dependencies are available in the supplementary section.

All end-uses aggregated allow to estimate hourly load patterns for regions and months and estimate future load shape changes by projected variation of climate conditions and/or uneven end-use demand increase. Finally, annual LDCs are built using the modelled monthly regional hourly patterns from weekday and weekend days. LDCs are commonly used within IAMs for capturing the dynamics between demand and supply in a simplified way. Within IMAGE, the ADVANCE LDCs are currently used for researching alternatives for balancing electricity demand and supply with the integration of VRE sources; however, their shapes do not change over time [15]. Therefore, for this research, annual regional LDCs were built for analyzing their change over time and for comparison purposes with the empirically-based LDCs from the ADVANCE project [14].

2.1. Input data

Empirically-based daily patterns found in the literature on hourly electricity demand by end-use were implemented to create the stylized demand patterns. However, publicly available measured data by end-use is very scarce. Few countries have collected these data and mostly from Europe and North America. As a result, each end-use pattern was calibrated with one set of empirically-based data. For end-use patterns without dependence on biophysical parameters, a set of empirically-based data was used to represent load demand in all regions. Table 2 depicts the data used for each end-use demand pattern.

The daily load shape of electric vehicles (EV) is expected to change over time, depending on the increasing penetration of EVs in the market and EV charging stations' technology development. Therefore, two charging demand pattern scenarios were implemented (Fig. 2), one based on currently used EV charging patterns (uncontrolled charging) and one based on the implementation of EV controlled chargers for charging the car at times of base-load, such as during midnight hours.

Table 2
End-use electricity demand data used.

End-Use demand	Data used	Location
Residential - Lighting and appliances	REMODECE campaign with 100 houses participating per country [24]. Appliances included: fridge-freezer, TV, dryer, washing machine, dishwasher, microwave oven.	12 European countries.
Residential- Heating	Based on Danish panel data measurements [25].	Denmark
Residential-Hot water	Pacific Northwest project [26] with 25 houses measured.	USA
Residential - Cooling	Pacific Northwest project with 415 air conditioners measurement in the residential sector [27].	USA- California
Residential - Cooking	UK measured data campaign on 250 households from 2010 to 2011 [28].	UK
All end-uses for the service sector	Extended database from 936 locations in USA in 2004 and it covers 16 reference types of commercial buildings for different end-uses [29]	USA
Transport – EV (Electric Vehicles)	The realistic scenario from Babrowski et al., 2014 [30] based on mobility surveyed data in Germany and reviewed with Schäuble et al., 2017 and Weiller, 2011 [31,32]. The scenario for controlled charging was inspired by a literature review on this topic [30,33–35].	Germany
Transport- Railways	Modelled data based on Netherland's railways demand patterns from the 1992 model by ECN institute (personal correspondence).	The Netherlands
Industries: food and paper	Panel data database from 2012. It covers 33 different electricity demand end-uses or industries with about 4500 representative customers [36].	Denmark
Industries: cement and steel	Cement and steel industries in China have continuous processes that require almost continuous daily load [37]. Therefore, a continuous daily load pattern was implemented. For their monthly variation, the panel data database from Denmark was implemented [36].	China/ Denmark

2.1.1. Occupancy profiles (Occ)

Occupancy profiles describe people's presence in the service or residential sector as a percentage of other activities. This information can be obtained from Time Use Surveys (TUS), which are questionnaires that gather how people spend their time daily. TUS are commonly used for modelling electricity demand because there is a demonstrated dependence between the timing of electricity demand and people's behavior [38–41]. Therefore, people's occupancy profile plays a vital role in modelling hourly demand for the service and residential sectors. Although TUS from three regions were available [42–44], the European database "HETUS" was the only one implemented in this study because of the distinction between weekday and weekend occupancy profiles and the distinction between sleeping and awaken at every hour of the day. It gathers data from 15 European countries between 1998 and 2004 [44]. For the residential sector, not only the location of people matters but also whether a person is sleeping or performing an activity in a building. Hence, four different profiles were derived from HETUS data for this sector: active occupancy profiles (only people awake are considered) and total occupancy profiles, both differentiating between weekdays and weekend days. The HETUS database was also implemented for deriving occupancy profiles of buildings in the service sector. Statistics related to the percentage of people working, sleeping, studying or eating at home or in the service sector were used [44]. Fig. 3

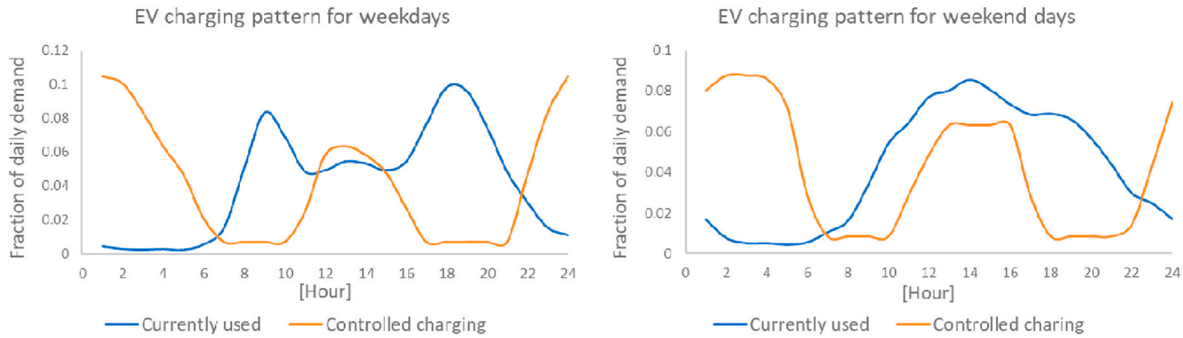

Fig. 2. *Electricity demand patterns implemented from EV charging for weekend (right side) and weekdays (left side). The blue curve corresponds to a currently used EV charging pattern* [30], *and the orange curve corresponds to a controlled charging pattern for demand valley filling* [30,33–35].

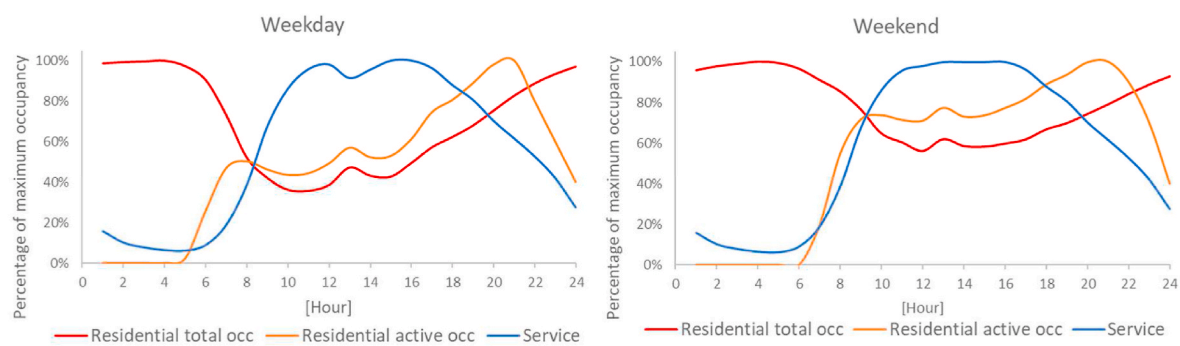

Fig. 3. Occupancy profiles (Occ) implemented in the model for residential and service sector for weekday and weekend days. Residential active occupancy profile (orange colour) exclude the percentage of people sleeping, while residential total occupancy profiles (red colour) include all people in this sector.

illustrates the occupancy profile obtained for the residential and service sector for weekday and weekend days.

2.1.2. *Biophysical parameters (BP)*

Two biophysical parameters are implemented in this research: temperature and irradiance. The relationship between temperature and electricity demand fluctuations has been primarily studied [45–50]. The temperature was found to directly influence the seasonal and/or daily electricity demand patterns for space cooling and heating in the service and the residential sector. A 24-h pattern of temperature variation for these end-uses for every month of a year was implemented. It was obtained by averaging five years of historical data of daily maximum and minimum temperature collected from the ISIMIP database from the climate model GFDL-ESM2M [51]. Then, a sinusoidal equation was applied for obtaining hourly day-variations. The hourly temperature data is then translated into cooling degree hours (CDH) and heating degree hours (HDH) which indicate the hourly variation of temperature compared to a reference temperature. An average reference temperature of 18 °C was applied both for cooling and heating. For seasonal variation over the years until 2100, monthly cooling degree days (CDD) and heating degree days (HDD) data previously calculated within IMAGE were used [52]. Concerning the influence of temperature on heating in the service sector, it was only found for the monthly variation; while for its hourly variation, the service heating daily patterns from the literature do not correlate with the mentioned parameters (building occupancy profile and the HDH). A reason for this is the so-called thermal inertia in buildings producing a time-lag and a reduction of heat flows between outdoor and indoor temperature, and it varies depending on the building materials used [53,54].

Temperature also influences electricity demand for water heating. However, this influence is related to seasonal variation and not to daily variation [55]. While seasonal water heating electricity demand is related to the temperature of water from pipelines, it mainly depends on ambient temperature [55]. Then, the seasonality factor was simplified to use a stylized function depending on HDD for the residential and service sector (the stylized functions are found in supplementary information).

The monthly variation of lighting demand from empirical data follows a trend that highly resembles the variation of hours with solar irradiance (i.e. daylength). There is an inverse relationship between monthly daylength variation and seasonal electricity demand for lighting. This is, in particular, the case for the residential sector [24,26,56]. However, this is not the only factor affecting monthly demand, national holidays and people behavior can also affect the monthly lighting demand in the residential sector. For the service sector, lighting demand seems very constant throughout the year, according to the reference data used from the USA [29]. Still, with some lower demand in summer times than in winter times. Therefore, a stylized function depending on daylength was implemented for this sector. Concerning lighting daily variation, for the service sector, it depends mainly on building occupancy and not so much on hourly irradiance [57,58]. However, for the residential sector, next to people's occupancy profile, hourly lighting demand resembles the sunrise and sunset hour variation. Therefore, a 24 h pattern of irradiance level varying per month is implemented as the biophysical parameter affecting electricity demand for residential lighting. This parameter was obtained by using a model that formulates daylight hours depending on latitude and day of the year [59]. Then, the daylight hours were distributed throughout the day using a sinus function centered around 12:30 h to obtain the day pattern of irradiance per region. Note that the stylized functions between the end-uses mentioned and the biophysical parameters are explained in detail in the supplementary section.

2.1.3. *Future sectoral electricity demand*

The hourly patterns are combined with projections of annual sectoral (end-use) electricity demand. For this, the SSP2 baseline scenario from the Shared Socioeconomic Pathways (SSPs) is implemented in the

IMAGE model [60]. The SSP2 is the "Middle of the road" pathway with medium assumptions for social, economic and technological parameters. In this scenario, the total energy demand continues to rise despite that the intensity of resource and energy use declines [61].

The annual contribution to the total electricity demand of the end-use sectors in the SSP2 scenario changes over time. They depend on drivers such as population growth, income, technology, and lifestyle assumptions which vary over time, depending on the SSP scenario implemented [52]. Fig. 4 illustrates how the contribution of sectoral demand to total demand changes between 2019 and 2070 under the SSP2 scenario with further granularity to cooling and heating in the residential and service sector. India is the region by far with the expected largest increase of total electricity demand within this period (indicated with the black dots in Fig. 4) and with the expected largest increase of cooling demand. For the USA and Western Europe, there is an increase in cooling demand of 81% and 96% compared to 2019 values, but it continues to have a low share of the total demand (3% for Western Europe in 2070). All regions in the graph, except for Western Europe and the USA, show an increase in the share of service demand. Especially for China and India, due to a significant increase in cooling demand. For most regions, the largest variation in end-use electricity demand comes from an expected rapid increase in electric vehicles (EV) charging demand, with the largest increase for Western Europe and Brazil.

3. Results

The end-use demand patterns are aggregated into total hourly electricity demand for each region implementing the SSP2 scenario. The hourly electricity demand patterns obtained vary over time and between regions. These variations are influenced by the climate-related parameters used (with regional, hourly and monthly variation) and annual end-use electricity demand modelled of the IMAGE-SSP2 scenario (with regional and annual variation). In this section the results are illustrated by first comparing the obtained regional modelled patterns with empirical electricity demand data. Second, the patterns variation over regions and time are illustrated and finally, the load duration curves and the changes over time are analyzed.

3.1. Comparison with empirical data

In this section, the aggregated yearly average patterns for weekday and weekend days are compared to empirical data for the regions where the data is available: Western Europe and Central Europe [62], USA [63] and Brazil [64]. Fig. 5 illustrates the comparison of the modelled and empirical average pattern for February for the region of Europe for weekday and weekend days. Modelled patterns for Western and Central Europe regions show the best fitting with the measured data. They depict well the pattern variation across regions, mainly for weekend days. For Europe, two load-peaks can be distinguished: the morning one is related to building heating demand in February; the evening one is related to peak demand from different end-uses in the residential sector. For Brazil, only one peak-load during evening hours can be distinguished in the empirical and modelled patterns. However, for all regions, the base-load from the modelled patterns is lower than the empirical patterns, and the base to peak-load ratio is lower for the measured patterns than the modelled ones. These differences could be related to the implementation of static patterns for most end-uses, especially for end-uses from the residential and service sectors that may vary across regions depending on people's culture (such as cooking and appliances use); this may affect the shape of the residential pattern for some of the regions. Furthermore, there is a scarcity of end-use empirical data, especially for the service sector, to validate the input data used and validate the end-use stylized functions obtained. This could mainly affect the peak to the base-load ratio for the service sector pattern. Still, the modelled patterns have their peak-load and base-load occurring at similar times as the empirical patterns and capture the general characteristics. In the supplementary section, a similar comparison is shown for September.

3.2. Patterns variation over region and time

Fig. 6 shows the regional variation of the average electricity demand day-pattern for the year 2019. It shows how the peak/base-load ratio can largely vary between regions. Typically, less developed regions depict a larger peak-load in evening hours related to the peak-load from the residential sector. This is shown for Eastern Africa in Fig. 5, but this can also be seen for other low-income regions such as Rest Central America, Rest South Asia, Northern and Western Africa. Furthermore, regions with a very high share of demand from the industry sector depict a high base-load. For example, this is the case for India, which has the highest base-load (72% of the peak-load), followed by China and South Africa.

The development of annual sectoral and end-use demand affects the shape of the hourly demand patterns over time. This is illustrated in Fig. 7 with the August weekday pattern variation between 2019 and 2070 and considering a currently used EV charging demand pattern. Further, for the year 2070, the patterns obtained with the implementation of the EV controlled charge demand patterns are also shown.

For the 2019 August demand pattern, all regions in Fig. 7 have a distinctive peak-load during evening hours caused by the evening peak-load from the residential sector (this is the least in Western Europe). By 2070, the changes in load shape are highly influenced by the rapid increase in demand from EV except for India. For this last one, the changes are related to a more rapid increase in service and residential sectors (especially from cooling) than in other sectors (Fig. 4) However, for Brazil, the Western Europe region, and to a lower extent also for China, it

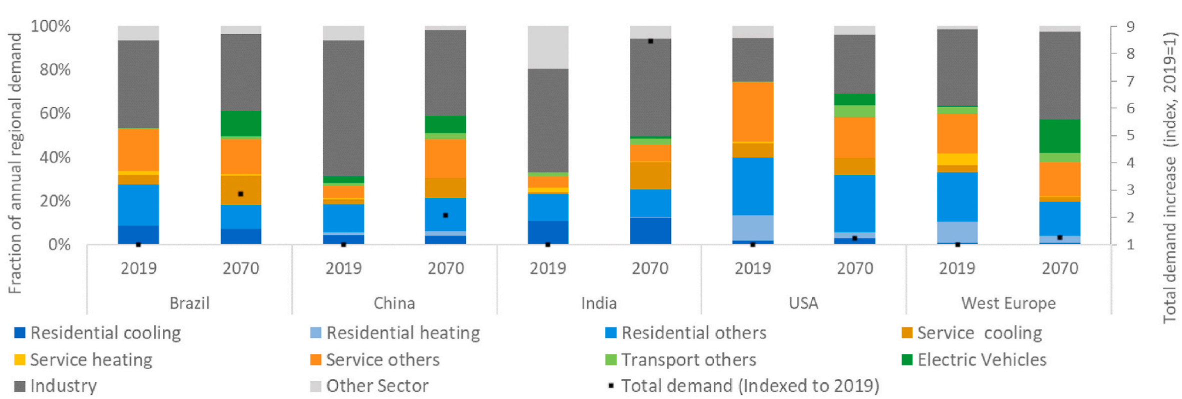

Fig. 4. Sectoral and end-use share of total electricity demand for 2019 and 2070 under the SSP2 scenario. The black dots represent the total demand indexed to the regional demand values for 2019 (right axis).

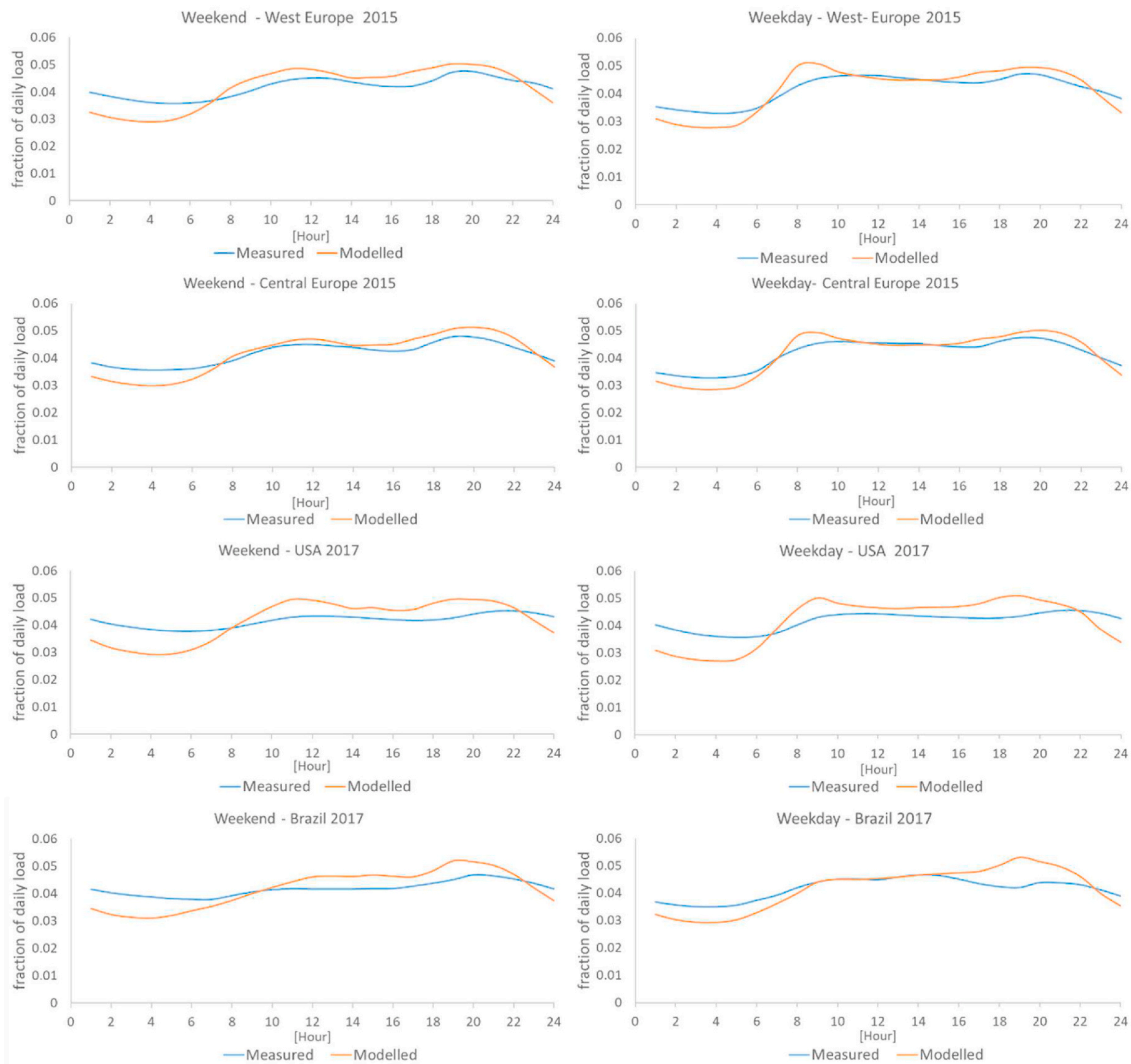

Fig. 5. *Comparison of measured and modelled hourly electricity demand patterns of the average weekday and weekend day for February for Europe, USA and Brazil. The years vary* per *graph depending on the availability of measured data* [62–64].

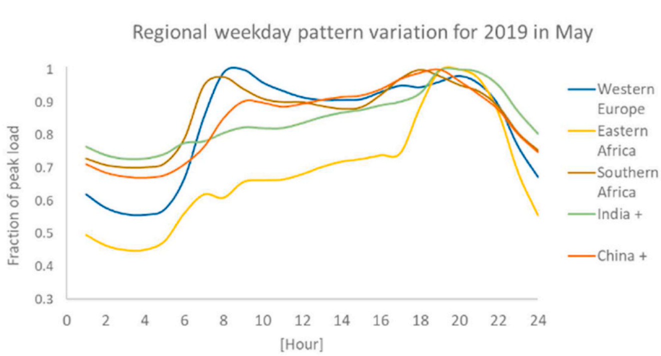

Fig. 6. Regional variation of the electricity demand weekday pattern in May 2019 for selected regions.

can be noted that future total demand patterns are sensitive to variations on the EV demand pattern. Total demand for 2070 with the implementation of currently used EV patterns (yellow curve) shows a morning and evening peak that coincides with those EV demand patterns. They occur in the morning when people arrive at the office and in the evening, coinciding with the residential sector's peak.

Under this scenario variant, the base-load and the peak-load difference increases towards 2070. However, when implementing the controlled EV charging pattern, the difference between the peak-load and base-load is lower for 2070 when compared to the 2019 curve. Furthermore, the peak-loads occur during daylight hours (around noon) and midnight hours. This result highlights the importance of implementing controlled charging stations that can respond to the generation pattern of VRE and that avoid charging the cars at hours of peak demand from sectors or end-uses that are more difficult to control, such as the residential sector.

3.3. Load duration curves

The LDCs were built using the monthly modelled patterns obtained of hourly electricity demand for weekdays and weekend (section 3.1 and 3.2). These patterns were then translated into one-year hourly data of electricity demand. Fig. 8 shows the modelled LDCs and a comparison with the empirical-based LDCs [14]. Note that the years of the modelled (blue line Fig. 8) and measured (orange line Fig. 8) LDCs vary depending on the availability of the empirically-based data as follows: Japan, India and USA LDCs correspond to the year 2010, Brazil LDCs are from the

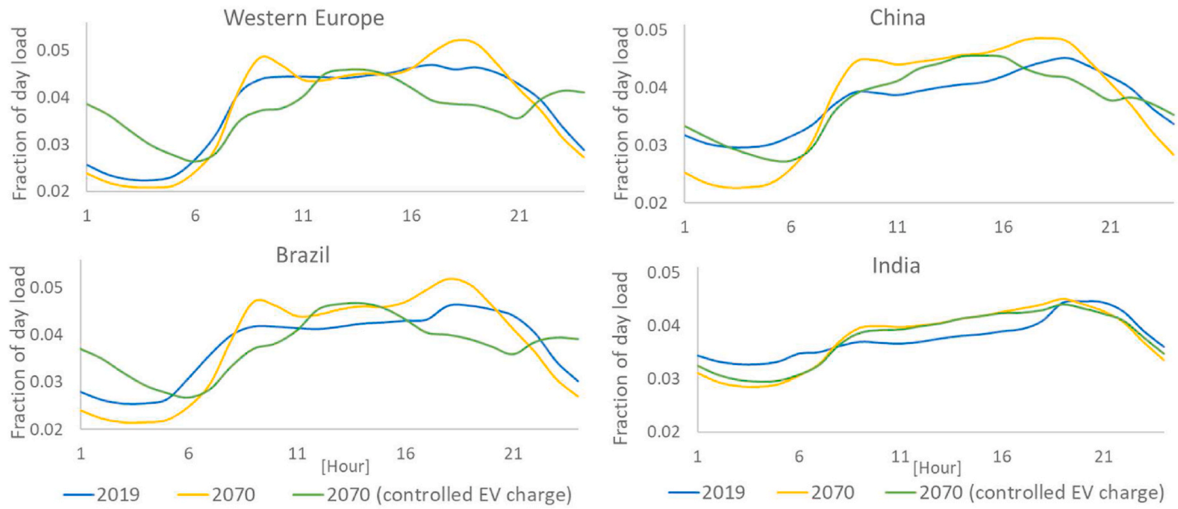

Fig. 7. Electricity demand annual pattern variation from the average weekday in August for the Western Europe region (top-left), China (top-right), Brazil (bottom-left), and India region (bottom-right). The green line represents the demand pattern for the year 2070 with the implementation of a controlled EV charging pattern.

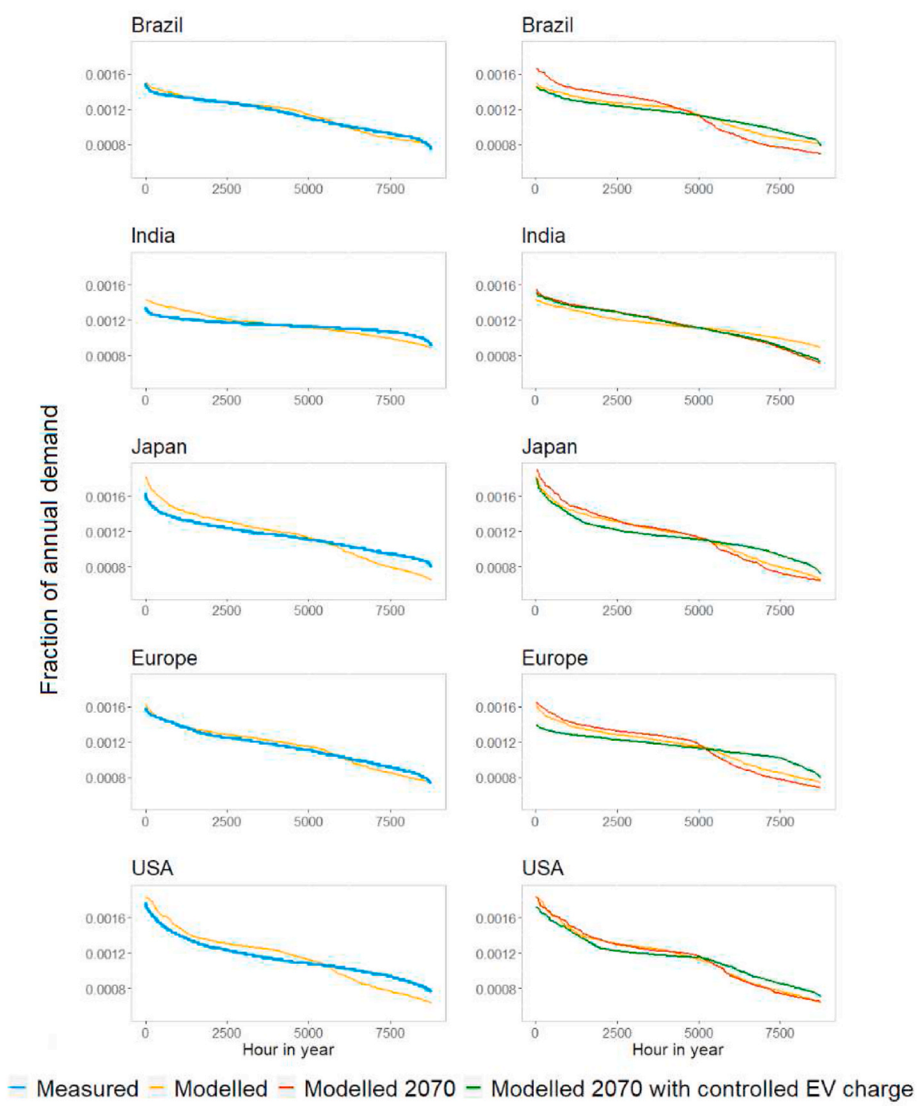

Fig. 8. Left side: Comparison of load duration curves between modelled and measured data extracted from Ref. [14]. Right side: Modelled present LDCs (orange line) compared to future LDCs for the year 2070 under two EV charging load scenarios. The red line represents a LDC for 2070 with uncontrolled charging based on present data. The green line represents a LDC for 2070 with a controlled charging pattern. The years of the modelled (blue line) and measured (orange line) LDCs vary depending on the availability of the empirically-based data as follows: Japan, India and USA LDCs correspond to the year 2010, Brazil LDCs are from the year 2009, and Europe LDCs are from the year 2006.

year 2009, and Europe LDCs are from the year 2006. Fig. 8 also shows 2070 LDCs under the SSP2 scenario using the current EV charging pattern (red line) and a controlled EV charging pattern (green line). The modelled LDCs reflect the general characteristics as observed in the empirically-based LDCs, especially for Brazil and Europe regions. However, for most regions, the modelled LDCs depict a larger base to peak-load ratio. For most regions, there is a visible drop in the LDC around the hour 5000 caused by the baseload from residential and service sector during late night hours. Examples for sectoral daily load patterns and LDCs are shown in the supplementary section.

Concerning the variation of LDCs over time, the regions of Europe, Brazil and Japan have more variation over time and show a larger sensitivity to changes in implementing the EV charging pattern for 2070. As expected, the implementation of EV controlled charging patterns has

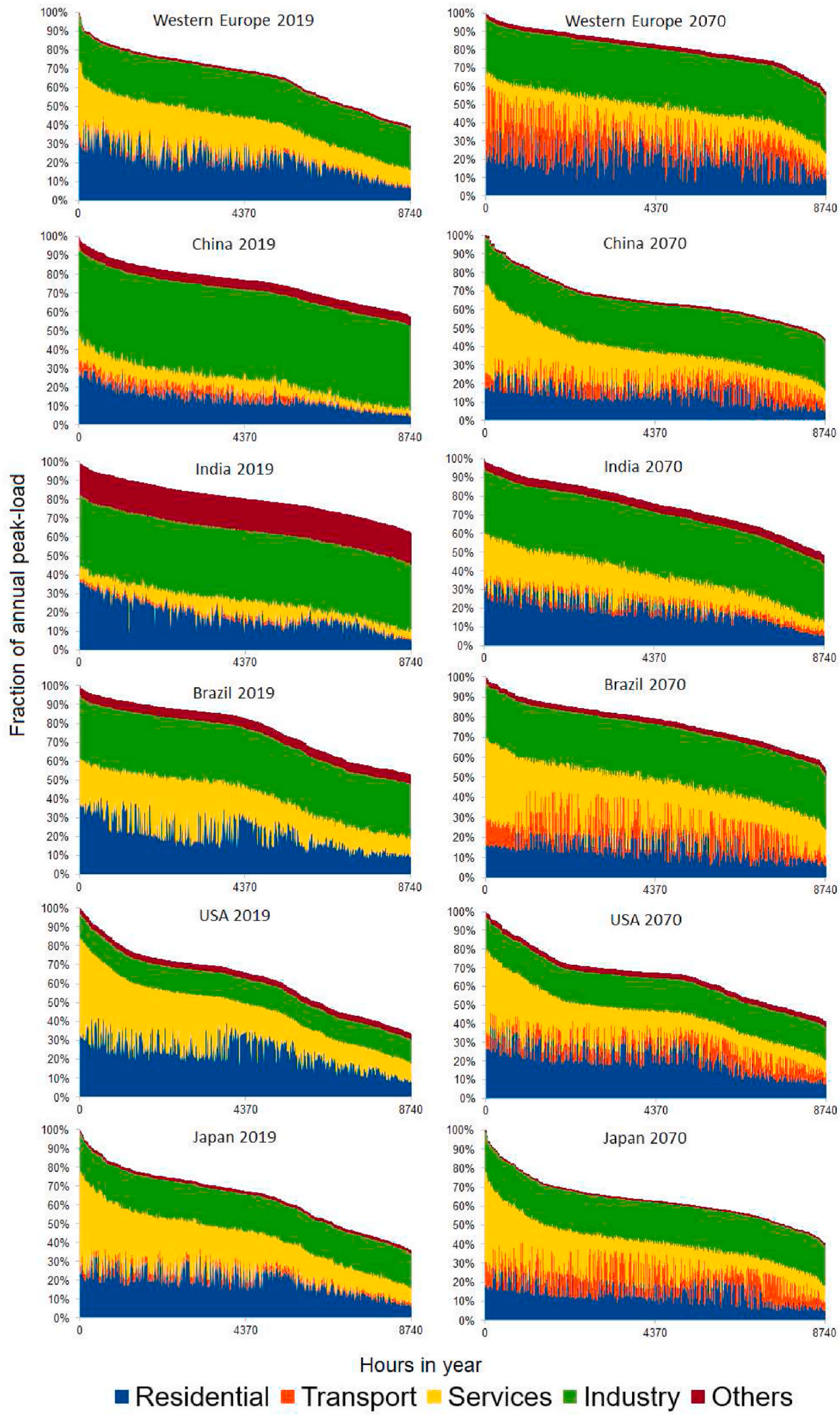

Fig. 9. Sectoral decomposition of load duration curves for selected regions. From left to right, the first corresponds to LDCs for 2019 with the current EV charging pattern. The second column corresponds to 2070 LDCs with the controlled EV charging pattern.

a flattening effect on the LDCs compared to the present (orange line) and 2070 LDCs with the currently used EV charging pattern (red line). The LDC from USA depicts very little variation over time as a result of relatively small changes in sectoral share in total electricity demand compared to other regions. Still, the low changes in its LDC are related to reducing the share of demand from the residential and service sector for 2070.

To better understand the effect of sectoral demand variation in the shape of LDCs, they have been decomposed on each of the electricity demand sectors considered. This is illustrated in Fig. 9 for the 2019 LDCs and the year 2070 with the implementation of a controlled EV charging pattern. For all regions in Fig. 9, the service and residential sectors' load curves contribute to a larger difference between peak and base-load due to their hourly and monthly patterns (non-stacked sectoral LDCs are shown at the end of the supplementary section). Additionally, the peaks from the service sector are at different times than the peaks from residential sector. The loads are larger during evening hours (after 16 o'clock) for the residential sector while for the service sector, the loads are larger before 17 o'clock. Furthermore, these peaks do not necessarily coincide with the largest load from total demand.

4. Conclusions and discussions

There are two visible trends in the electricity sector: an increase in demand for electricity and a rapid growth in the electricity production from renewable energy sources. This poses a significant challenge: how can the variable supply from solar and wind match demand? To solve this issue, detailed knowledge is required about both the timing of electricity supply and the timing of electricity demand. In this paper, it is discussed how the timing of demand is different between regions and how it may change over time. This research aims to generate hourly patterns of electricity demand with a global scope that vary over time and across the different world regions for long-term exploratory scenario analyses.

4.1. Discussions

Because of the scarcity of empirical end-use hourly demand data, simplifications were made that could exclude details on regional variations. For example, one set of hourly demand patterns was implemented for all regions for some end-uses with a strong cultural component, such as the use of appliances and demand for cooking.

Since we limit ourselves to empirically observed demand patterns, not all relevant options that can have an influence on load demand patterns are implemented:

Specific demand-side management policies were not included in the assessment, and they may affect the shape of daily demand, especially in the future. For example, heat-pumps usage whose demand patterns are different from traditional electric heaters [65] and they have the potential to respond to changes in electricity supply or price if coupled with thermal storage. Furthermore, improvements in building insulation and its implementation at a large scale could also influence the daily pattern of heating demand. These two examples are essential strategies for decarbonizing heating demand in buildings [66].

The effect of new technologies (with different load shapes) penetrating the market cannot be captured with models based on regression analysis or historical load curves [67]. In this regard, a future improvement would be the modelling of the EV-to-grid concept for supporting the balance of supply and demand with high penetration of solar and wind sources [68]. This concept together with power to fuels (for example, for producing hydrogen and synthetic fuels) can be important strategies for supporting the penetration of VREs.

5. Conclusions

Based on our results, the following conclusions were obtained:

The model can capture the general features of the empirical load demand patterns. Aggregated weekday and weekend total hourly electricity demand patterns were compared to empirical data for four regions. The peak-load and the base-load of the modelled and the empirical patterns occur at a similar time. They can capture differences between the regions observed in the empirical data, especially for weekend days. For instance, in February, both the model and empirical patterns depict two peak-loads for Europe; while for Brazil, one peak-load is distinguished during evening hours. Overall, the regions of Western and Central Europe show the best fitting. However, the difference between peak and base-load is larger for our modelled patterns over the regions compared.

The model captures LDCs and daily load shape variations across regions, with less developed regions typically depicting a larger peak-load in evening hours. These are the Eastern Africa, Rest of Central America, Rest of South Asia, Northern and Western Africa regions. They all depict a larger share of the residential sector demand, and a lower demand share from industry and/or service sector compared to developed regions. Further, regions with the highest share of demand from the industry sector depict higher base-load, such as India, China and South Africa. In our view, the variation of demand patterns across regions is well captured.

For all regions and under the middle of the road scenario, small changes can be expected in the shape of LDCs and daily load of electricity demand over time. The changes are more visible for the daily load patterns compared to the LDCs, especially for the second half of this century. For most regions, such as Brazil, Western Europe and China, the changes mainly result from the rapid increase in demand from electric vehicles (EV) charging. For India, the changes of demand pattern over time are mostly determined by a more rapid increase in service sector demand compared to the other sectors. The variation of hourly demand over region and time is captured by first the uneven end-use annual increase in demand. Secondly, it is influenced by the projected variation of biophysical parameters that affect the modelled end-use patterns.

Future electricity demand day patterns and load duration curves (LDCs) are expected to have a large sensitivity to load shape variations from electric vehicle daily charging patterns. The LDCs and load patterns modelled for 2050 and 2070 with the current used EV charging patterns depict a larger difference between the peak and base-load than the 2019 situation. However, a flattening of future load curves can be expected by implementing controlled EV charging technologies. This highlights the importance of implementing a bottom-up approach for modelling future long-term electricity demand load. Furthermore, this method gives insight into the effect of policies that incentivize the implementation of new technologies that can respond to electricity pricing and production. They would enable a reciprocal relationship between electricity production, pricing and demand and then facilitate the penetration of variable renewable energies (VREs).

For future research, the electricity hourly demand patterns modelled in this research can be combined with potential hourly VRE estimations. This will help IAMs to better capture the dynamics between electricity demand and (VRE) electricity supply within different mitigation strategies in IAMs. For instance, the correlation between solar supply and the projected air-conditioning demand increase in the India's service sector is not captured with the currently used static residual load duration curves. The dynamic load patterns developed here are the first step into improving the estimations of electricity production and storage capacity needed, dispatch order, residual load after supplying the demand with VRE sources, and investments needed for balancing supply and demand under different demand scenarios. All this facilitates the analysis and implementation of international climate policy for the energy sector.

Credit author statement

Harmen-Sytze de Boer, Detlef van Vuuren and Victhalia Zapata developed the idea, Victhalia Zapata designed the experiments and wrote the article. Raul Maicas contributed with the LDCs decomposition. All authors discussed the results and contributed to the manuscript.

Code availability

PBL holds the proprietary rights of the IMAGE computer code, extensive documentation is provided (https://models.pbl.nl/image/index.php/Welcome_to_IMAGE_3.0_Documentation).

Credit author statement

VZ, HSdB and DvV developed the idea, VZ developed the experiments and wrote the article. RMM contributed with the LDCs decomposition. All authors discussed the results and contributed to the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Modelled end-use electricity demand obtained and total hourly electricity demand per region are provided in the following link: https://data.mendeley.com/datasets/pmd2dchk44/1.

Acknowledgements

The paper benefitted from funding from the European Research Council under grant ERC-CoG PICASSO (n° 819566).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2022.124741.

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